ChatGPT Practical – Labs 1 to 6

LAB 1: Introduction to ChatGPT

Prompt: Explain what ChatGPT is, including its origin, working, capabilities, and major real-world applications. Use simple language and give examples from education, healthcare, business, and creativity.

Output: ChatGPT is an AI chatbot developed by OpenAI. It is based on the GPT family of language models and is designed to understand natural-language questions and instructions.

ChatGPT works using a Transformer-based architecture. It processes the input, understands the context, and generates a response based on patterns learned during training.

Capabilities:
• Answering questions
• Explaining concepts
• Summarizing information
• Writing and rewriting text
• Generating and debugging code
• Creating ideas and quizzes

Applications:
• Education: Study material and explanations
• Healthcare: General information and documentation assistance
• Business: Emails, reports, and customer support
• Creativity: Stories, ideas, and content generation

LAB 2: Types of Prompts and Prompt Engineering

Prompt 1 – Zero-Shot

Prompt: Explain what a Large Language Model is.

Output: A Large Language Model (LLM) is an AI system trained on a large amount of text to understand and generate human-like language. It can answer questions, summarize text, translate languages, write content, and assist with coding.

Prompt 2 – One-Shot

Prompt: A Foundation Model in AI refers to a model like GPT-3, which is trained on a large dataset and can be adapted to various tasks. Explain what BERT is in this context.

Output: BERT (Bidirectional Encoder Representations from Transformers) is a Transformer-based language model developed by Google. It is trained on large amounts of text and understands the context of words by considering surrounding words. BERT can be used for tasks such as text classification, question answering, and sentiment analysis.

Prompt 3 – Few-Shot

Prompt: Foundation Models such as GPT-3 are used for natural language processing, while models like DALL-E are used for image generation. How are Foundation Models used in the field of robotics?

Output: Foundation Models can be used in robotics to help robots understand instructions, recognize objects, plan actions, and interact with humans. They can combine information from language, images, and sensor data to support intelligent robotic behavior.

Prompt 4 – Instructional

Prompt: Explain Artificial Intelligence in simple language. Define AI, explain how it works, list three real-world applications, and mention two advantages and two limitations. Use headings and bullet points. Keep the explanation suitable for engineering students.

Output: Artificial Intelligence (AI) is a technology that enables machines to perform tasks that normally require human intelligence.

How AI Works:
• Uses data to learn patterns.
• Uses algorithms to process information.
• Produces predictions, decisions, or responses.

Applications:
• Healthcare
• Banking
• Transportation

Advantages:
• Saves time
• Automates repetitive tasks

Limitations:
• Can produce incorrect results
• Requires quality data

Prompt 5 – Interrogative

Prompt: What is Artificial Intelligence, how does it work, and what are its major applications in real life?

Output: Artificial Intelligence is a technology that enables machines to perform tasks that normally require human intelligence. It works using data, algorithms, and machine learning techniques to identify patterns and produce results. AI is used in healthcare, education, banking, transportation, entertainment, and many other fields.

LAB 3: Precision Prompting for Information Extraction

Prompt: Extract the following information from this text and return it only in JSON format:

"Rahul Sharma is a Python developer. His email is rahul@gmail.com and his phone number is 9876543210. He has skills in Python, Java, SQL and Machine Learning."

Extract: Name, Job, Email, Phone, Skills

```
{
  "Name": "Rahul Sharma",
  "Job": "Python developer",
  "Email": "rahul@gmail.com",
```

```
  "Phone": "9876543210",
  "Skills": ["Python", "Java", "SQL", "Machine Learning"]
}
```

LAB 4: Summarization and Text Transformation

Prompt 1 – Summarization

Prompt: Summarize the following paragraph in 5 bullet points:

"Artificial Intelligence is a technology that enables computers to perform tasks that normally require human intelligence. AI is used in healthcare, education, banking, transportation and entertainment. Machine learning allows systems to learn from data and improve their performance."

Output: • Artificial Intelligence enables computers to perform tasks requiring human intelligence.
• AI is used in many different industries.
• Major applications include healthcare, education, banking, transportation, and entertainment.
• Machine Learning is an important part of AI.
• Machine Learning helps systems learn from data and improve their performance.

Prompt 2 – Tone Transformation

Prompt: Rewrite the following message in a formal and polite tone:

"Hi Sir, I am not able to attend the lab today. Please give me the instructions for the experiment."

Output: Dear Sir,
I am unable to attend the laboratory session today. Kindly provide me with the instructions for the experiment so that I can complete the work.
Thank you.

LAB 5: ChatGPT in Code Generation and Debugging

Prompt: Find and correct the errors in this Python program:

```
num = input("Enter a number: ")
if num % 2 == 0
   print("Even")
else:
   print("Odd")
```

Explain what errors were present and provide the corrected code.

Output: Errors:
1. input() returns a string, so it must be converted to an integer using int().
2. A colon : is missing after the if condition.

Corrected Code:

```
num = int(input("Enter a number: "))

if num % 2 == 0:
```

```
        print("Even")
    else:
        print("Odd")
```

Output: Example Output:
Enter a number: 8
Even

LAB 6: Domain-Specific Applications – Educational Quiz

Prompt: Create a quiz on Artificial Intelligence for engineering students. Generate 10 multiple-choice questions, 4 options for each question, mark the correct answer, and include questions of easy, medium and difficult levels.

Output: 1. [Easy] What is Artificial Intelligence?
A. Technology that enables machines to perform intelligent tasks
B. A programming language
C. A database system
D. A computer network
Correct Answer: A

2. [Easy] Which is a subfield of AI?
A. Machine Learning
B. HTML
C. SQL
D. Networking
Correct Answer: A

3. [Easy] Which is an AI application in healthcare?
A. Medical image analysis
B. Keyboard manufacturing
C. File storage
D. Voltage control
Correct Answer: A

4. [Medium] What is the purpose of training data in Machine Learning?
A. To help the model learn patterns
B. To increase processor speed
C. To replace the operating system
D. To remove algorithms
Correct Answer: A

5. [Medium] Which task is an example of classification?
A. Predicting whether an email is spam
B. Predicting house price
C. Compressing a video
D. Storing a file
Correct Answer: A

6. [Medium] Why is overfitting undesirable?

A. The model may perform well on training data but poorly on new data
B. The model cannot be trained
C. The dataset becomes larger
D. The computer loses its operating system
Correct Answer: A

7. [Medium] What is the role of weights in a neural network?
A. They determine the strength of connections between neurons
B. They store the operating system
C. They control monitor resolution
D. They replace training data
Correct Answer: A

8. [Difficult] What is the main purpose of the attention mechanism in Transformer models?
A. To focus on relevant parts of the input
B. To increase computer storage
C. To replace the CPU
D. To compress files
Correct Answer: A

9. [Difficult] Which learning method uses examples with known target outputs?
A. Supervised Learning
B. Unsupervised Learning
C. Random Learning
D. Manual Learning
Correct Answer: A

10. [Difficult] A classifier makes 90 correct predictions out of 100. What is its accuracy?
A. 90%
B. 80%
C. 10%
D. 9%
Correct Answer: A

End of Practical Record